Traditional Knowledge

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Traditional Knowledge

1. Purpose and Question of the Investigation

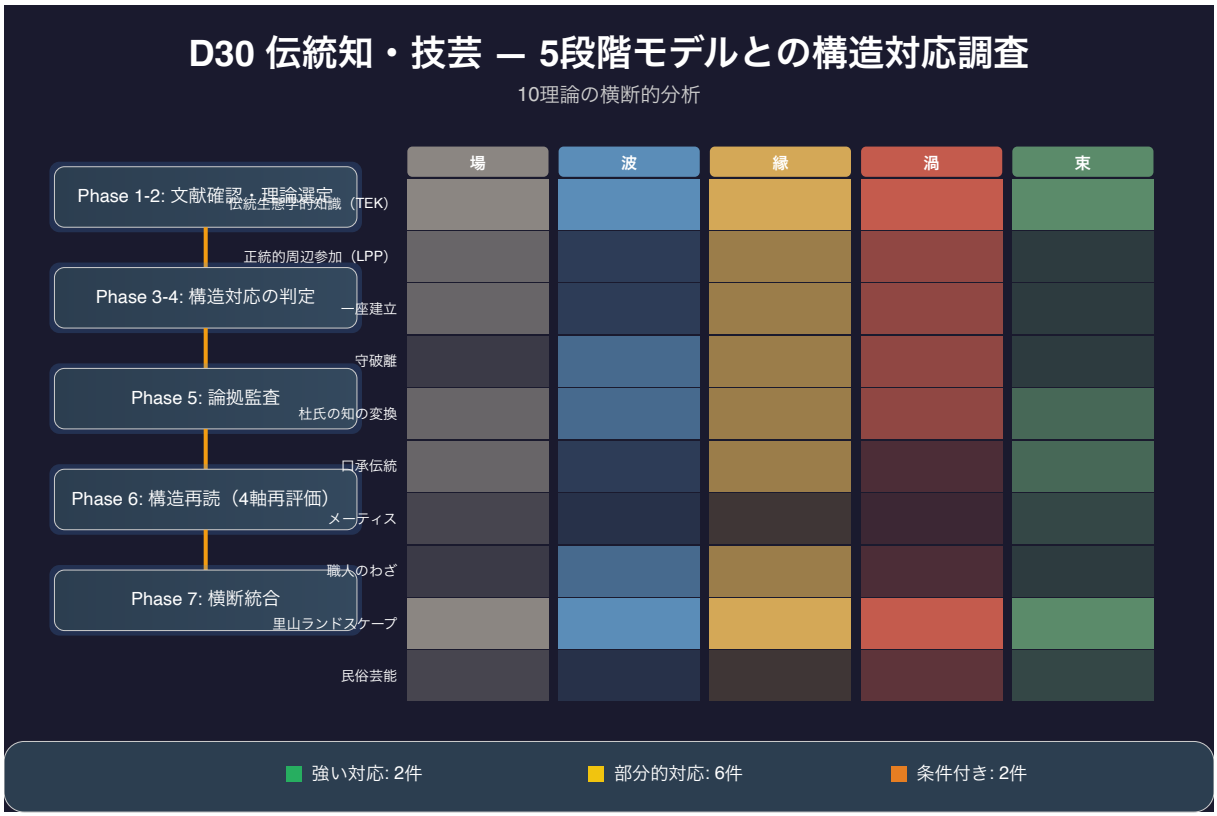

Figure 1: Traditional Knowledge —

This report was prepared to answer the following question:

> Do the theories in this academic domain correspond structurally to the five-stage model (Field ( ) -> Wave ( ) -> Edge ( ) -> Vortex ( ) -> Bundle ( ))?

Traditional knowledge and arts comprise systems of knowledge, practice, and belief accumulated and transmitted across generations, encompassing the acquisition of forms, the transformation of tacit knowledge, and the emergence of collective intelligence. Their structural similarity to the five stages is especially pronounced at the levels of Edge ( ), where meaning is woven within relationships, and Bundle ( ), where patterns settle into institutions and lineages of transmission.

This report presents the findings of an investigation; it is not a document written to justify any particular theory. Areas of strong correspondence and areas that require reservation are described with the same analytical temperature.

2. Method of Investigation

Scope and Selection Criteria

In this domain, 11 theories and practical models were selected for evaluation. Selection was based on three criteria: (1) explicit treatment of processes of generation or transformation; (2) the possibility of structural analysis on the basis of primary texts or well-established secondary literature; and (3) the capacity to compare distinct lineages within the domain.

Evaluation Procedure

For each theory, I first explain what the theory describes, and then examine where correspondences with the five stages can be identified. Judgments of correspondence were made comprehensively on the basis of textual concepts, concrete examples, connections between stages, the presence or absence of circulation from Bundle ( ) to the next Field ( ), and the risk of overinterpretation.

Criteria of Judgment

- **Strong structural correspondence**: the core concepts of the theory correspond concretely to multiple stages of the five-stage model, and the basis for that judgment can be explained in the text
- **Conditional correspondence**: correspondence can be indicated, but interpretive reservations remain in mapping the stages
- **No confirmed correspondence**: partial resemblance exists, but the overall five-stage structure cannot be read without strain

Limits of the Investigation

This investigation does not offer an exhaustive review of the complete works associated with each theory. Instead, it focuses on the core concepts of major texts and extracts structure at a comparable level of granularity. As a result, some internal details of the theories and some historiographical disputes could not be treated fully.

3. Overview of the Model

The five-stage model describes the process of creation in the following five phases.

Stage	Definition
Field	An undifferentiated state. It refers to initial conditions in which neither direction nor structure has yet been fixed.
Wave	A stage of exploration in which multiple directions diverge and differences are amplified.
Edge	A stage of tension in which opposing elements coexist without immediately converging on either side.
Vortex	A stage in which a new coherence or order arises from within that tension.
Bundle	A stage in which form becomes determinate and stabilizes as a reusable structure.

This model is not a one-way line; it includes circulation in which Bundle ( ) becomes the condition for the next Field ( ). Moreover, the five stages are not best understood as strictly partitioned boxes, but as a continuous spectrum. The model includes the hypothesis that the same structure can recur across different scales, from momentary thought to institution-building and long-term historical change.

If a sense of incongruity remains at this point, that itself is an important perspective for the present investigation. Readers may continue while retaining that sense of incongruity.

Figure 2: Traditional Knowledge — ×5

4. Findings: Overall Picture

List of Theories Evaluated

Theory	Proponent(s)	Structural correspondence
TEK: Traditional Ecological Knowledge	Berkes, Colding & Folke (2000)	Strong
LPP: Legitimate Peripheral Participation	Lave & Wenger (1991)	Strong
Ichiza Konryu: co-creation of the gathering	Zeami's *Fushikaden*, Yamanoue Soji's *Yamanoue Sojiki*	Strong
Shu-Ha-Ri: preserving, breaking, and transcending form	Geido traditions (tea, martial arts), Zeami-related transmission texts	Strong
Toji skill transmission: tacit knowledge in sake brewing	Asahi Shuzo case (Dassai), Nonaka-related SECI literature	Strong
Oral Tradition (Ong/Vansina)	Ong (1982), Vansina (1985)	Strong
Folk knowledge / metis (Scott/Sillitoe)	Scott (1998), Sillitoe (1998)	Strong
Craftsmanship (Ikuta Kumiko / Sennett)	Ikuta Kumiko (1987), Sennett (2008)	Strong
Satoyama landscape	Takeuchi et al. (2003, 2012), SATOYAMA Initiative (CBD COP10, 2010)	Strong
Transmission processes in folk performing arts	Honda Yasuji (folk performing arts studies), Japanese folklore studies	Conditional
Jo-Ha-Kyu: theory of Noh and performance		Strong

Of the 11 items evaluated, 10 showed strong structural correspondence, 1 showed conditional correspondence, and 0 showed no confirmed correspondence. Overall, this domain exhibits correspondence with the five-stage model at a relatively high density, but the mode of correspondence is not uniform. Some theories describe the conditions of Field ( ) in thick detail, while others offer precise accounts of the tensions of Edge ( ) or the fixation characteristic of Bundle ( ).

What matters most is not the high rate of correspondence as such, but which theory describes which stage through which vocabulary. Even when treating the same phenomenon of "creation," this domain illuminates different moments of the five stages through distinct layers such as body, institution, time, perception, practice, and symbol.

5. Findings: Examination of Major Theories

5.1 TEK: Traditional Ecological Knowledge

"TEK: Traditional Ecological Knowledge," as articulated by Berkes, Colding & Folke (2000), is defined as a cumulative complex of knowledge, practice, and belief that is culturally transmitted across generations and evolves through adaptive processes. It describes adaptive management

practices that include multispecies management, resource rotation, succession management, and the management of landscape patchiness. Huntington (2000) points out practical obstacles to documenting TEK because it is centered on oral transmission. IPBES organizes ILK (Indigenous and Local Knowledge) as a dynamic and integrated system of knowledge, practice, and belief.

The TEK cycle of "observation -> interpretation -> response -> institutionalization" corresponds structurally to the five stages. Especially noteworthy is the way "ecological surprise" in TEK functions as an ecological version of rupture or discrepancy. Deviation from expected seasonal patterns is recognized by the community as an anomaly, and rather than being dismissed, it is taken up as a question. At the stage of Edge ( ), experienced members compare signs and reinterpret causal relations; this is the organization of relational knowledge, a process in which meaning arises not within isolated individual cognition but within the web of relations constituting the community.

TEK occupies an interdisciplinary space spanning conservation ecology, environmental governance, and Indigenous studies. Since the 1990s it has been linked, especially by Berkes and Folke, to resilience theory, and its possible integration with scientific knowledge has been widely discussed. In recent years, it has also entered the policy sphere through IPBES. Its use in the present project is limited to the indication of structural similarity rather than the extraction or repurposing of knowledge content, but if made public, it would be desirable to specify the scope and purpose of reference in consideration of FPIC principles.

5.2 LPP: Legitimate Peripheral Participation

"LPP: Legitimate Peripheral Participation," proposed by Lave & Wenger (1991), understands learning as "situated activity" and defines its central characteristic as legitimate peripheral participation. It describes learning as a trajectory through which newcomers deepen their participation in the practices of a community. Observation of the Vai/Gola tailoring apprenticeship system in West Africa served as the point of departure for concept formation. Wenger (1998) later refined the idea of "Communities of Practice," presenting four constitutive concepts: negotiation of meaning, participation, reification, and identity.

The participation trajectory described by LPP corresponds structurally to the five stages. The community preexists as Field ( ); with the entry of newcomers, the difference between periphery and center rises into view as Wave ( ). Through legitimate access, relationships with old-timers are woven as Edge ( ); skills and identity are then formed together as Vortex ( ); and eventually the learner moves into the position of reproducing the community itself as Bundle ( ). The stage of Edge ( ) is especially concrete in LPP, where it appears as both the legitimacy of access and the organization of relations. The result is a clear structure in which knowledge is generated within relationships rather than being merely transmitted as information.

LPP stands at the intersection of education, anthropology, and organization theory. As a theory of situated learning, it has been widely influential since the 1990s, and Wenger's concept of the community of practice has become broadly established in knowledge management and organizational learning. In the context of D30, it should be positioned not in relation to formal educational institutions, but as a mechanism of informal learning in apprenticeship systems and traditional communities. It is complementary to the SECI model in management studies, but it operates on a different axis because its focus is not knowledge conversion but participation and becoming a member.

5.3 Ichiza Konryu: co-creation of the gathering

"Ichiza Konryu: co-creation of the gathering," found in Zeami's *Fushikaden* and Yamanoue Soji's *Yamanoue Sojiki*, is presented in *Fushikaden* through a passage that identifies the successful

formation of the gathering with “shunin aikei,” thereby positioning the formation of a gathering as a core concept of performance. *Yamanoue Sojiki* includes wording to the effect that, from entering the roji until leaving it, host and guest participate in an encounter that occurs only once in a lifetime, emphasizing the formation of a singular, unrepeatable gathering.

The process of Ichiza Konryu corresponds structurally to the sequence Field ( ) -> Wave ( ) -> Edge ( ) -> Vortex ( ) -> Bundle ( ). The tearoom, utensils, and arrangement compose the Field ( ); entry into the room generates an extraordinary tension as Wave ( ); through exchanges mediated by host and guest gestures, silence, and minimal language, Edge ( ) takes shape; from this emerges a felt unity, “this gathering has truly come into being,” as Vortex ( ). The experience is then remembered and normatively sedimented as a single encounter, becoming Bundle ( ) that remains as structure for the next Field ( ). The transition from Edge ( ) to Vortex ( ), that is, the emergence of collective unity from relational exchange, is the structure most vividly displayed by Ichiza Konryu.

Ichiza Konryu is a central concept in Japanese geido traditions, especially Noh and tea. In Zeami’s theory of Noh, it appears as resonance between performer and audience; in tea, as co-creation between host and guest. Although the textual corpus overlaps with D28, which treats Zeami’s flower, D30 positions Ichiza Konryu instead as a process of emergence in collective knowledge. Recent empirical studies suggest that “co-creation of the place” can be achieved even beyond prior experience, implying that Ichiza Konryu may possess a universal structure not limited to highly trained experts.

5.4 Shu-Ha-Ri: preserving, breaking, and transcending form

“Shu-Ha-Ri: preserving, breaking, and transcending form,” found in Japanese geido traditions such as tea, martial arts, and flower arrangement, as well as in texts related to Zeami, names stages of training: shu is the phase of faithfully practicing inherited forms; ha is the phase of applying those forms and introducing personal ingenuity; ri is the phase of departing from the form to attain an independent level of mastery. Etymologically, the concept is often said to trace back to Zeami’s theory of performance, though the historical moment and context in which the term “Shu-Ha-Ri” itself was clearly formalized require closer terminological scrutiny.

In this theory, the five-stage correspondence appears in the movement from formed preconditions, through amplified differences and organized relations, to the emergence of a new coherence and its eventual stabilization.

Shu-Ha-Ri is a core concept of Japanese geido traditions and is widely known as a general account of training across tea, martial arts, flower arrangement, and Noh. In recent years it has also been repurposed in business and software development, including agile practice. Although part of its content overlaps with entries on Zeami in aesthetics and art, D30 focuses not on individual artistic excellence but on Shu-Ha-Ri as a mechanism for the transmission of knowledge.

5.5 Toji skill transmission: tacit knowledge in sake brewing

“Toji skill transmission: tacit knowledge in sake brewing,” discussed through the Asahi Shuzo case (Dassai) and in relation to Nonaka’s SECI model, concerns the abolition of the traditional toji system at Asahi Shuzo and the sharing of brewing information among employees through datafication. Various indicators, including temperature, sugar content, and acidity, were measured and analyzed to manage the production process, establishing a system that fed results back into the next round of brewing. The case is widely cited in knowledge-management theory as a process of converting tacit knowledge into explicit and then organizational knowledge, and it can be read as a concrete instance of the SECI model.

The transformation of tacit knowledge into explicit and then organizational knowledge in the transmission of toji skills corresponds clearly to Field ( ) -> Wave ( ) -> Edge ( ) -> Vortex ( ) -> Bundle ( ). When numerical measurement is introduced into the toji's sensory judgment, a discrepancy emerges between what is grasped only vaguely and what is represented numerically; this appears as Wave ( ). Through datafication and sharing, tacit knowledge is translated into transmissible form as Edge ( ). It stabilizes as a standardized production process, or Vortex ( ), and is finally bundled as organizational knowledge together with cyclical improvement, becoming Bundle ( ). A distinctive feature is that the stage of Edge ( ) is made visible in the concrete mediating boundary of "datafication."

The toji system is a traditional Japanese mechanism for transmitting brewing skill, and the institution of the toji living in the brewery while making sake has a history of several centuries. The case is widely cited in knowledge management and management studies as a symbol of institutional transformation. While there is overlap with the SECI model in management studies, D30 positions the case from the domain-specific angle of the transformation of traditional knowledge.

5.6 (Oral Tradition - Ong/Vansina)

" (Oral Tradition - Ong/Vansina)," represented by Ong (1982) and Vansina (1985), draws on Walter Ong's *Orality and Literacy* (1982), which analyzes the structural differences between oral and literate cultures. Ong describes as characteristic of oral culture its reliance on formulaic expression, additive and aggregative modes of thought, dialogical and agonistic tone, and homeostatic renewal. Jan Vansina's *Oral Tradition as History* (1985) established a methodology for using oral tradition as a historical source. It classifies oral tradition into testimony, narrative, commentary, formulaic phrases, and related types, while analyzing mechanisms of transformation in the process of transmission.

The generative process of oral tradition corresponds structurally to the five stages. Within a shared lifeworld as Field ( ), events or anomalies become differentiated as things that must be told, appearing as Wave ( ). In the face-to-face performance between teller and audience, the narrative is co-constructed as Edge ( ). It crystallizes into formulaic expressions such as songs, proverbs, and myths as Vortex ( ), and is transmitted across generations as tradition, becoming Bundle ( ). Especially important is the stage of Edge ( ): oral performance is not one-way transmission but an interaction in which the audience's responses alter the telling. This interactivity is the structural feature that distinguishes oral tradition from mere information transfer.

Research on oral tradition has two great classics in Ong's *Orality and Literacy* (1982) and Vansina's *Oral Tradition as History* (1985). The former has been influential in media theory and literacy studies; the latter in African history and historical methodology. Both are well-established scholarly works that continue to be widely cited. In the context of D30, they are positioned as accounts of the mechanisms by which knowledge is transmitted before writing.

5.7 folk knowledge/ (Scott/Sillitoe)

" folk knowledge/ (Scott/Sillitoe)," associated with Scott (1998) and Sillitoe (1998), draws first on James C. Scott's *Seeing Like a State* (1998), which analyzes the structure by which high-modernist state planning fails through its disregard for metis, that is, practical wisdom. Scott defines metis as locally specific, experience-based practical knowledge and contrasts it with techne, systematic technical knowledge. Sillitoe (1998), working in development anthropology, examines the relationship between local or Indigenous knowledge and global scientific knowledge, describing local knowledge as situation-dependent and context-embedded.

Folk knowledge and metis display structural similarity to the five stages, but their point of connection is distinctive. The most direct connection lies in the stance of taking up what would otherwise be dismissed as an error or discrepancy and treating it as a question. The high-modernist state Scott describes "sees" society through maps, statistics, and plans, yet in doing so it discards locally specific differences that do not fit within that system of vision. Metis is the knowledge system that retrieves those discarded differences and responds to them practically. In addition, the structure by which adaptive response is generated at the point of contact between official knowledge (techne) and unofficial knowledge (metis) can be read as a political and social version of Edge ( ).

Scott's *Seeing Like a State* is a classic in political science, anthropology, and development studies, and the concept of metis is very widely cited. Sillitoe's work established the position of local knowledge in development anthropology. Both are well-established academic sources. In the context of D30, they are complementary to TEK, but whereas TEK focuses on ecological knowledge, metis concerns a broader practical wisdom including agriculture, architecture, and social organization.

5.8 Craftsmanship (Ikuta Kumiko / Sennett)

"Craftsmanship," based on Ikuta Kumiko (1987) and Sennett (2008), begins from Ikuta Kumiko's *Waza kara shiru* (1987), which analyzes the process of acquiring waza in Japanese traditional performance and craft. Waza is not acquired through transmission as information, but by coming to understand through bodily engagement. The book describes a process involving imitation, repetition, failure, and the integration of bodily sensation. Richard Sennett's *The Craftsman* (2008) argues that handwork is fundamental to the human condition. It presents resistance from materials as the core of skill development and describes hand-brain dialogue as the site of creation. It also refers to the notion of 10,000 hours of training.

The acquisition of craftsmanship corresponds structurally to the five stages. Within the Field ( ) composed by workshop, material, and tools, the fact that materials do not behave as expected appears as resistance, which rises as Wave ( ). This is the material manifestation of the discrepancy between prediction and reality. Rather than forcing the material into submission, the craftsperson responds to it through hand-brain dialogue, thereby entering Edge ( ). Out of this dialogue, skill crystallizes as Vortex ( ), and is preserved institutionally as craft tradition in the form of Bundle ( ). The stage of Edge ( ) is particularly clear in Sennett's concrete formulation of "hand-brain dialogue," which points to the process by which skill is woven at the boundary between intention and material.

What craftsmanship teaches here is how discrepancy extends into material practice, and how relational formation takes bodily form. The stance of treating the resistance of materials not as noise to be discarded but as a question to be taken up is itself a paradigmatic craft practice. Sennett's emphasis on "10,000 hours of training" also connects to Shu-Ha-Ri in showing a common structure: quantitative accumulation becomes the condition for qualitative transformation. Ikuta's claim that one begins from imitation resonates as well with LPP's account of peripheral participation.

5.9 (SATOYAMA) Landscape

" (SATOYAMA) landscape," discussed by Takeuchi et al. (2003, 2012) and the SATOYAMA Initiative (CBD COP10, 2010), refers to a socio-ecological production landscape formed in the contact zone between human settlement (sato) and woodland or mountain (yama) in Japan. Takeuchi et al. systematized the concept within landscape ecology. The SATOYAMA Initiative was launched at CBD COP10 in Nagoya in 2010 and promotes the sustainable use of socio-ecological production landscapes and seascapes (SEPLs) within an international framework. It is

positioned as a model for the coexistence of human activity and biodiversity.

The structural similarity between the Satoyama landscape and the five stages is especially clear. The contact zone itself between sato, the human domain, and yama, the natural domain, is Edge ( ), and management of that boundary through disturbance produces a mosaic landscape. Within the socio-ecological system as Field ( ), seasonal fluctuation and human disturbance generate differences as Wave ( ); at the boundary between sato and yama, multiple elements interact as Edge ( ); the maintained landscape mosaic stabilizes as Vortex ( ); and management institutions, common rights, and ecological calendars are preserved across generations as Bundle ( ). What is particularly striking is that the "edge" of Satoyama is, both semantically and structurally, the boundary itself. It is therefore one of the most intuitive examples of the concept of Edge ( ) in the five-stage model.

Satoyama was originally a landscape concept specific to Japan, but after the launch of the SATOYAMA Initiative in 2010 it was incorporated into international biodiversity-conservation frameworks. It has been widely studied in landscape ecology, conservation ecology, and environmental policy. In recent years, the decline of Satoyama through the abandonment of management has become a major issue, making it important as an empirical example of the effects produced by abandoning the management of edges. Although there is overlap with agriculture and ecology, D30 focuses on Satoyama as a body of traditional knowledge concerning landscape management.

5.10 Transmission processes in folk performing arts

"The transmission processes of folk performing arts," associated with Honda Yasuji and Japanese folklore studies, concern collective performances handed down in local communities across Japan, including ritual performance, kagura, Bon dance, and lion dance. Honda Yasuji's multi-volume *Nihon no minzoku geino* provided the foundation for systematic documentation and classification. The transmission of folk performing arts occurs not primarily through texts but through repeated collective performance. Preparation for annual festivals, including practice, role allocation, and costume preparation, is the actual site of transmission, and performance itself simultaneously functions as an act of transmission.

The transmission processes of folk performing arts correspond to the five stages. Grounded in the community's annual cycle as Field ( ), the arrival of the festival season triggers a shift from the ordinary to the extraordinary as Wave ( ). Through intergenerational practice and preparation, techniques are transmitted in Edge ( ); they are gathered into collective performance as Vortex ( ); and then structurally passed on to the next year and the next generation as festival institutions, repertoires, and roles, becoming Bundle ( ). What is especially noteworthy is that the annual festival cycle provides a visible example of the return from Bundle ( ) to Field ( ): the prior year's bundle of institutions, repertories, and roles becomes part of the next year's field, and transmission proceeds in a spiral.

The study of folk performing arts is an important branch of Japanese folklore studies, and the work of Honda Yasuji laid much of its foundation. Festivals, kagura, and Bon dances across Japan are also protected as elements of intangible cultural heritage. Although there is overlap with performance theory in aesthetics and art, D30 focuses on the mechanism of transmission, positioning these practices not in terms of artistic value but as processes of intergenerational knowledge transfer. Because the literature is mainly in Japanese, there are limits to international verification at this level.

5.11 Jo-Ha-Kyu: theory of Noh and performance

"Jo-Ha-Kyu: theory of Noh and performance" appears in texts related to Zeami and in theories of Noh. Discussions of *Shugyokutokka* have sometimes attached to Jo-Ha-Kyu an interpretation

that aligns it with a general theory of motion, though it has also been suggested that the text may contain relatively little substantial content as a theory of training or performance.

In this theory as well, correspondence with the five stages appears in the movement from formed preconditions, through amplified differences and organized relations, to the emergence of a new coherence and its stabilization.

The significance of this theory lies in showing which moments of the five stages are described in greatest density through the vocabulary specific to the domain.

6. Cross-Cutting Patterns

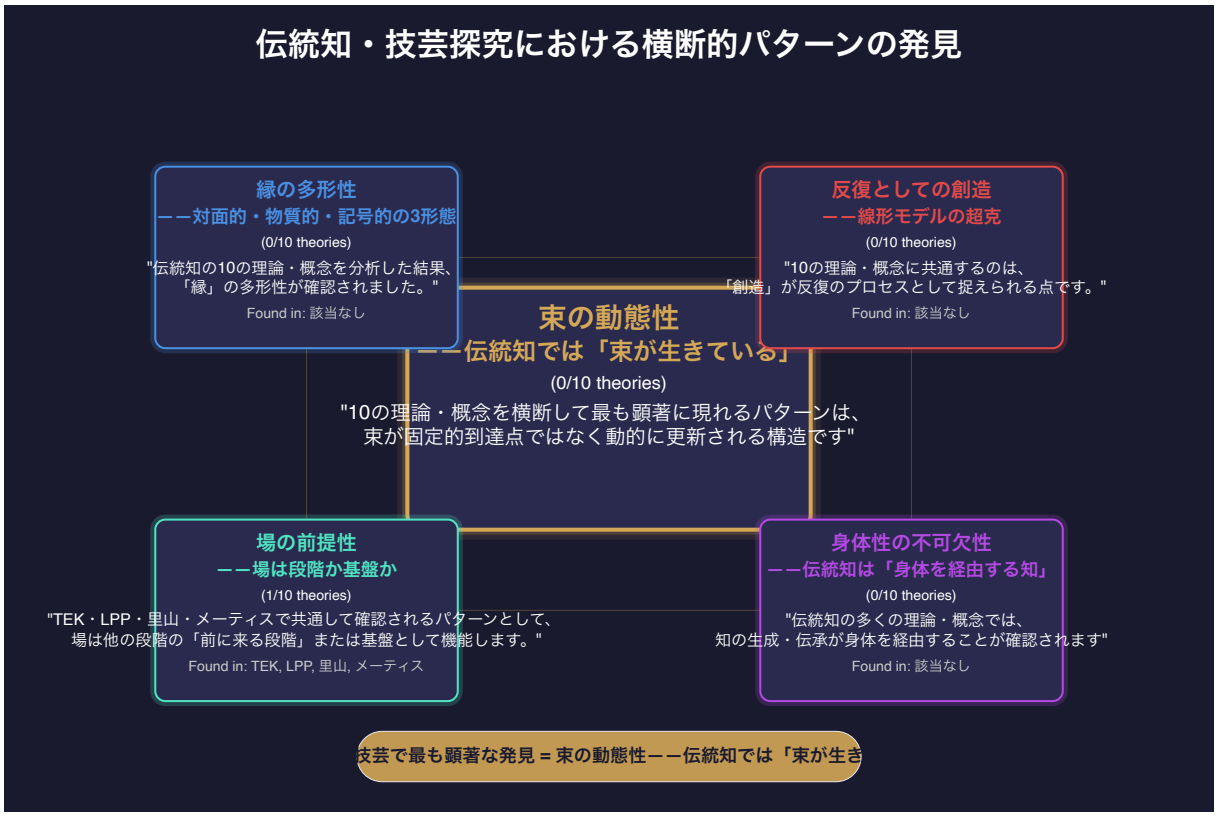

Figure 3: Traditional Knowledge —

First, as shown by TEK: Traditional Ecological Knowledge, LPP: Legitimate Peripheral Participation, and Ichiza Konryu: co-creation of the gathering, this domain describes difference and conflict not merely as obstacles but as conditions under which a new order can arise. This means that, among the five stages, the transition from Edge ( ) to Vortex ( ) can be observed here with particular richness.

Second, what Shu-Ha-Ri: preserving, breaking, and transcending form, Toji skill transmission: tacit knowledge in sake brewing, and Oral Tradition (Ong/Vansina) share is that generation is not a one-time event, but includes a cycle of forming preconditions, amplifying tension, producing local ordering, stabilizing it, and then moving into the next reorganization. This confirms that the five-stage model must be read not as a static classification chart but as a temporal movement.

At the same time, the transmission processes of folk performing arts also show that correspondence with the five stages does not arise automatically. A mere match in the number of stages or in the proximity of terms is insufficient; it is necessary to examine case by case which concepts carry the turning points and at what level structure takes shape.

7. Unresolved Questions

1. To what extent can the transmission processes of folk performing arts be read as a genuinely five-stage description?
2. To what extent can the structures confirmed in this domain be extended from the explanation of individual cases to a more general model of generation? The conditions under which comparability can be preserved without erasing the specificity of the objects still require close scrutiny.
3. Should the five stages be read as a linear progression, or rather as a circulation involving multiple temporal layers and repetitions? In particular, theories differ in how explicitly they describe the return from Bundle ( ) to the next Field ( ).
4. What becomes visible, and what is lost, when vocabulary specific to this domain is translated into the general vocabulary of the five stages? The boundary between what is translatable and what is not is itself a major issue for future comparative investigation.

8. Conclusion

This reexamination of the domain confirmed strong structural correspondence in 10 of the 11 theories considered, and conditional correspondence in 1. In particular, TEK: Traditional Ecological Knowledge, LPP: Legitimate Peripheral Participation, and Ichiza Konryu: co-creation of the gathering describe in rich detail the movement from Field ( ) to Edge ( ), or from Edge ( ) to Vortex ( ), and can therefore be positioned as central evidence within this domain.

At the same time, the findings of this domain do not support the claim that the same five stages simply appear unchanged in every theory. The stages that become most visible differ from one theory to another, and some cases require reservations in mapping the stages. What matters, therefore, is not to generalize directly from a high rate of correspondence, but to continue comparing which moments are described through which vocabularies.

In that sense, this domain does not merely support the validity of the five-stage model in a one-way manner; it is also a domain that imposes conditions on how the model should be read. The very coexistence of theories that correspond thickly and theories that require reservation constitutes part of the value of the comparative investigation.